

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem

COURSE: Number Theory & Discrete Mathematics Application

CLASS: S3 CSE B

STUDENT NAME: Joel Duke

ROLL NO: 51

ROLE / FUNCTION: Mathematical Research & Problem Formulation

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working traffic signal model and an interactive digital web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3 min, 5 min, 7 min) align to create a continuous 'Green Wave' for vehicles at Minute 23.

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
 2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
 3. To model traffic signal timing cycles (3 min, 5 min, 7 min) as modular congruences.
 4. To build a physical working model and an interactive web simulator for signal synchronization.
 5. To provide clear visual and audio feedback when a candidate time satisfies all signal modulo conditions.
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3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 minutes respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 minutes ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 minutes ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 minutes ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M_1 = 35$, $M_2 = 21$, $M_3 = 15$.

3. Modular Inverses:

- $35 \cdot y \equiv 1 \pmod{3} \Rightarrow 2 \cdot y \equiv 1 \pmod{3} \Rightarrow y = 2$
- $21 \cdot y \equiv 1 \pmod{5} \Rightarrow 1 \cdot y \equiv 1 \pmod{5} \Rightarrow y = 1$
- $15 \cdot y \equiv 1 \pmod{7} \Rightarrow 1 \cdot y \equiv 1 \pmod{7} \Rightarrow y = 1$

4. Compute x:

- $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
- $x = 140 + 63 + 30 = 233$
- $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

The unique green-wave solution within $[0, 104]$ is Minute 23.

4. DIGITAL IMPLEMENTATION & USER CONTROLS

The web application models the traffic signal system digitally using React, TypeScript, and Framer Motion.





Interactive User Controls:

1. Candidate Timer Wheel: Adjusts the candidate time value x (e.g. set to 23).
2. Intersection Signal Monitors (3 min, 5 min, 7 min): Shows real-time remainder values ($x \bmod m$) and green light indicator status.
3. 'Evaluate Synchronization' Button: Tests whether all 3 intersections display green lights simultaneously.
4. Green-Wave Status Panel: Displays full green-wave confirmation when all 3 congruences are satisfied ($x = 23$).

5. SOURCE CODE IMPLEMENTATION (ALGORITHM PAGE)

The following complete TypeScript module (src/utlis/crt.ts) implements the Extended Euclidean Algorithm and Modular Inverses to solve the CRT system:

```
export function getRemainder(value: number, modulus: number): number {
    return ((value % modulus) + modulus) % modulus;
}

function extendedGcd(a: number, b: number): [number, number, number] {
    if (b === 0) return [a, 1, 0];
    const [g, x1, y1] = extendedGcd(b, a % b);
    return [g, y1, x1 - Math.floor(a / b) * y1];
}

function modInverse(a: number, m: number): number {
    const [, x] = extendedGcd(getRemainder(a, m), m);
    return getRemainder(x, m);
}

/** Solves  $x \equiv \text{remainders}[i] \pmod{\text{moduli}[i]}$  for pairwise-coprime moduli */
export function solveCRT(moduli: number[], remainders: number[]): number {
    const product = moduli.reduce((acc, m) => acc * m, 1);
    let result = 0;
    for (let i = 0; i < moduli.length; i++) {
        const mi = moduli[i];
        const ri = getRemainder(remainders[i], mi);
        const partial = product / mi;
```

```
        const inverse = modInverse(partial, mi);
        result += ri * partial * inverse;
    }
    return getRemainder(result, product);
}

export function checkRemainders(x: number, moduli: number[], remainders: number[]): boolean[] {
    return moduli.map((m, i) => getRemainder(x, m) === getRemainder(remainders[i], m));
}
```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3 min, 5 min, 7 min), the project demonstrates how non-zero remainder offsets determine the exact minute (Minute 23) required to establish a continuous 'Green Wave' for vehicles.

Both the physical working model and the interactive web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Joel Duke

Class: S3 CSE B

Roll No: 51

Role: Mathematical Research & Problem Formulation

Contribution Summary:

I researched the mathematical foundation of linear congruences. I formulated the traffic signal green-wave equations for 3, 5, and 7-minute cycles and calculated the unique solution bounds modulo 105.